The period-one locus in $\mathrm{GL}(2, \mathbb{Z})$: an intrinsic characterization, and a selection that sees only the trace

26 August 2026

Abstract

Let $A \in \mathrm{GL}(2, \mathbb{Z})$ be hyperbolic with non-negative entries. We show that the continued-fraction expansion of its dominant eigenvalue is purely periodic of period one *if and only if* $\det A = -1$, in which case $\operatorname{tr} A = m \geq 1$ and the eigenvalue is the m-th metallic mean $\lambda_m = \frac{1}{2}(m + \sqrt{m^2 + 4})$. The locus so described is thus not a family chosen to make a theorem true: it is cut out by a determinant condition, and the classification along it is by $(\operatorname{tr}, \det)$ alone.

Two refinements make the statement usable. First, the companion matrix X_m realizes each attained trace but is not always the unique $\mathrm{GL}(2, \mathbb{Z})$-class doing so; we compute the number of primitive $\mathrm{GL}(2, \mathbb{Z})$-classes of binary quadratic forms of discriminant $m^2 + 4$ to be $1, 1, 1, 1, 1, 2, 1, 1, 2, 2, 1$ for $m = 1, \ldots, 11$, so the class is unique exactly for $m \leq 5$ and $m = 6$ is the first repetition. Second — and this is the point of the paper — the invariant that singles out the golden member is insensitive to that ambiguity: for *every* A on the locus with $\operatorname{tr} A = m$, Cayley–Hamilton gives the integer-matrix identity

$$A^2 - I \;=\; mA,$$

so that $\det(A^2 - I) = -m^2$ and, because A is unimodular, the mapping torus of A^2 has first-homology torsion exactly $\mathbb{Z}/m \oplus \mathbb{Z}/m$ — a knot complement precisely when $m = 1$. The selection therefore operates on the whole period-one locus and not merely on the chosen representatives, which removes the objection a reader raises first. We also price the period-one restriction against the period-two relaxation — there the torsion is $\mathbb{Z}/\gcd(a, b) \oplus \mathbb{Z}/\operatorname{lcm}(a, b)$ and the only knot complement is again the golden one — and we state precisely what is *not* obtained: the passage from a substitution rule to an oriented mapping torus requires typed data that no result here supplies.

1 Introduction

1.1 The objection this paper answers

A theorem of the form "*inside the family $\mathcal{F}$, the object x_0 is the unique member with property P*" is worth exactly as much as the description of $\mathcal{F}$ is worth. If $\mathcal{F}$ was assembled by listing the objects that resemble x_0, the theorem is a tautology wearing a quantifier. The sharpest form of the objection is that the family admits no description independent of its conclusion.

We answer that form of it here for one specific family, and we are explicit about the weaker forms we do not answer.

The family is the set of hyperbolic $A \in \mathrm{GL}(2, \mathbb{Z})$ with non-negative entries whose dominant eigenvalue has a purely periodic continued fraction of period one. This description mentions no distinguished member. Theorem 2.2 shows it coincides with the determinant condition $\det A = -1$; Theorem 4.1 shows that the invariant which distinguishes one member from the rest depends only on the trace, hence is constant on each of the finitely many $\mathrm{GL}(2, \mathbb{Z})$-classes sharing a trace, hence cannot be an artifact of which representative one writes down.

1.2 Results

Throughout, $\lambda_m = \frac{1}{2}\big(m + \sqrt{m^2+4}\big)$ denotes the m-th *metallic mean*, the positive root of $x^2 - mx - 1 = 0$; $\lambda_1 = \varphi$ is the golden mean and $\lambda_2 = 1 + \sqrt{2}$ the silver.

- **Theorem 2.2** (characterization). Period one $\iff$ $\det A = -1$; then $\operatorname{tr} A = m \geq 1$, every $m \geq 1$ occurs, and the eigenvalue is λ_m. The classification is by $(\operatorname{tr}, \det)$ alone.
- **Theorem 3.2** (how far the representative is canonical). The number of primitive $\mathrm{GL}(2, \mathbb{Z})$-classes at discriminant $m^2 + 4$ is 1 for $m \leq 5$, and $m = 6$ is the first trace carrying two. So X_m *is* the unique class for $m \leq 5$, and the ambiguity is a genuine phenomenon from $m = 6$ on rather than a formal caution.
- **Theorem 4.1** (the selection sees only the trace). For every A on the locus with $\operatorname{tr} A = m$ one has $A^2 - I = mA$, hence $\det(A^2 - I) = -m^2$ and torsion $\mathbb{Z}/m \oplus \mathbb{Z}/m$ in the mapping torus of A^2, a knot complement iff $m = 1$. The identity fails off the locus, so it tests the locus rather than an accident of 2×2 matrices.
- **Proposition 5.1** (the restriction, priced). Period-two words are orientation-preserving already; the period-one family is the diagonal of the period-two family; and the only knot complement in the strictly larger two-parameter family is still the golden one. Against the one relaxation we tested, the restriction costs nothing.

1.3 What is not claimed

We claim nothing about how one arrives at such an A. In particular:

(a) **No canonical passage from a substitution to a 3-manifold.** Squaring a determinant-(-1) incidence matrix produces an orientation-preserving map, but the assignment of letters to Dehn twists, the choice of puncture, the orientation, and the mapping-torus construction itself are additional typed data. Nothing here supplies them, and we do not treat the resulting 3-manifold as determined by the combinatorics.

(b) **No claim that the seed is forced.** That some minimal-description principle picks out this substitution rather than another is not addressed and is not used.

(c) **No uniqueness of the conjugacy class for $m \geq 6$**, by Theorem 3.2; the results that survive that ambiguity are exactly the ones stated in terms of $(\operatorname{tr}, \det)$.

(d) **No physical interpretation.** None is offered and none is needed for any statement in this paper.

1.4 Relation to what is classical

Galois' theorem — a quadratic irrational has a purely periodic continued fraction exactly when it is *reduced*, i.e. $\alpha > 1$ and $-1 < \bar{\alpha} < 0$ — is classical [1, 2]. The metallic means and their period-one expansions are elementary and often observed. The classification of integral binary quadratic forms up to $\mathrm{GL}(2, \mathbb{Z})$, and the Latimer–MacDuffee correspondence between conjugacy classes of integral matrices and ideal classes [3, 4], are likewise standard.

What we believe is not already recorded in this combination is the packaging: the *equivalence* of the period-one condition with a determinant condition on the locus (Theorem 2.2), the identification of $m = 6$ as the exact threshold where the companion representative stops being canonical (Theorem 3.2), and above all the observation that the homological selection is a function of the trace alone (Theorem 4.1), which is what makes the selection survive the class ambiguity. We make no priority claim on the classical ingredients.

2 The characterization

Definition 2.1. For $m \geq 1$ put

$$X_m = \begin{pmatrix} m & 1 \\ 1 & 0 \end{pmatrix}, \qquad \phi_m = X_m^2 = \begin{pmatrix} m^2+1 & m \\ m & 1 \end{pmatrix},$$

so $\det X_m = -1$, $\operatorname{tr} X_m = m$, $\det \phi_m = +1$. Write M_A for the mapping torus of the once-punctured torus under a homeomorphism inducing A on H_1.

Theorem 2.2 (the locus is a determinant condition)**.** *Let $A \in \mathrm{GL}(2,\mathbb{Z})$ be hyperbolic with non-negative entries. The continued fraction of its dominant eigenvalue is purely periodic of period one if and only if* $\det A = -1$. *In that case* $\operatorname{tr} A = m$ *for some* $m \geq 1$, *the eigenvalue is* λ_m, *every* $m \geq 1$ *is attained, and the classification along the locus is by* $(\operatorname{tr}, \det)$ *alone.*

Proof. ($\Rightarrow$) Suppose the dominant eigenvalue x has expansion $[m; m, m, \dots]$ of period one, $m \geq 1$. Then $x = m + 1/x$, i.e. $x^2 - mx - 1 = 0$. The companion matrix of $x^2 - mx - 1$ is X_m, with $\det X_m = -1$ and $\operatorname{tr} X_m = m$. Since A and X_m have the same characteristic polynomial, $\det A = -1$ and $\operatorname{tr} A = m$.

($\Leftarrow$) Suppose A is hyperbolic with non-negative entries and $\det A = -1$. Its characteristic polynomial is $x^2 - tx - 1$ with $t = \operatorname{tr} A$. Non-negativity and hyperbolicity force $t \geq 1$: the entries are non-negative so $t \geq 0$, and $t = 0$ gives $x^2 = 1$, not hyperbolic. The dominant root is $\lambda_t > 1$ and satisfies $x = t + 1/x$, so its expansion is $[t; t, t, \dots]$, purely periodic of period one.

For attainment, X_m has non-negative entries, determinant -1 and trace m for each $m \geq 1$. For the final clause, the characteristic polynomial of a 2×2 matrix is determined by $(\operatorname{tr}, \det)$, and on this locus det is constant; so the trace determines the polynomial, the eigenvalue, and the expansion. Distinct m give distinct traces and hence distinct conjugacy classes. The converse is not injective on classes — see Theorem 3.2 — only on traces. ☐

Remark 2.3. The ($\Leftarrow$) direction is Galois' reducedness criterion in disguise: $\lambda_t > 1$ and its conjugate $\bar\lambda_t = \frac{1}{2}(t - \sqrt{t^2+4})$ satisfies $-1 < \bar\lambda_t < 0$ precisely because $0 < 4t$, so λ_t is reduced and its expansion is purely periodic. Period *one* is then the statement that the expansion's single partial quotient is $\lfloor \lambda_t \rfloor = t$.

3 How canonical is the representative?

Theorem 2.2 classifies by $(\operatorname{tr}, \det)$, not by conjugacy class. These differ, and it is worth knowing exactly where.

Scope 3.1 (Latimer–MacDuffee). Conjugacy classes in $\mathrm{GL}(2,\mathbb{Z})$ of matrices with a given irreducible characteristic polynomial correspond to classes of ideals in the associated order [3, 4]. For $x^2 - mx - 1$ the discriminant is m^2+4, and the count is the primitive $\mathrm{GL}(2,\mathbb{Z})$-class number of that discriminant. It need not be 1.

Theorem 3.2 (the threshold is $m = 6$)**.** *Let $h^+(m)$ denote the number of* $\mathrm{GL}(2,\mathbb{Z})$*-classes of primitive integral binary quadratic forms of discriminant* m^2+4. *Then*

$$\big(h^+(m)\big)_{m=1}^{11} = (1,1,1,1,1,2,1,1,2,2,1).$$

Consequently X_m is the unique *class of trace m on the locus for $1 \le m \le 5$, and $m = 6$ is the first trace at which it is not.*

Computation. By reduction theory for indefinite forms, each class contains reduced forms (a, b, c) with $b^2 - 4ac = D$, $0 < b < \sqrt{D}$ and $\sqrt{D} - b < 2|a| < \sqrt{D} + b$, and the reduced forms in a class constitute a single ρ-cycle; $\mathrm{GL}(2, \mathbb{Z})$-equivalence, as opposed to $\mathrm{SL}(2, \mathbb{Z})$-equivalence, additionally identifies (a, b, c) with $(-a, b, -c)$. Enumerating reduced primitive forms and grouping them into ρ-cycles modulo that identification gives the stated counts. The computation is reproduced, with controls, by the script named in Appendix A. □

Remark 3.3 (a recorded disagreement)**.** An independent count communicated to us reports 2 primitive classes at $m = 12$, where our enumeration gives 3. We have not resolved the discrepancy and therefore state Theorem 3.2 only for $m \le 11$. Nothing in this paper depends on $m \ge 12$: the threshold claim concerns the *first* repetition, which occurs at $m = 6$, and both counts agree on $1 \le m \le 11$. We flag it rather than suppress it.

Remark 3.4 (the trap this exposes)**.** Restricting to *primitive* forms is not a technicality. Counting all forms, imprimitive ones included, changes the answer — content is a $\mathrm{GL}(2, \mathbb{Z})$-invariant, so imprimitive forms contribute spurious classes — and in particular alters the count at $m = 4$, which would move the apparent threshold. The counts above are primitive throughout.

4 The selection sees only the trace

This is the section the paper exists for. Theorem 3.2 says the representative is not canonical past $m = 5$. If the invariant that distinguishes the golden member were a function of the representative, the distinction would be suspect. It is not.

Theorem 4.1 (trace-only selection)**.** *Let $A \in \mathrm{GL}(2, \mathbb{Z})$ with $\det A = -1$ and $\operatorname{tr} A = m \ge 1$. Then, as an identity of integer matrices,*

$$A^2 - I \;=\; mA.$$

Consequently $\det(A^2 - I) = m^2 \det A = -m^2$ *exactly, the mapping torus M_{A^2} has*

$$H_1(M_{A^2}; \mathbb{Z}) \;\cong\; \mathbb{Z} \oplus \mathbb{Z}/m \oplus \mathbb{Z}/m,$$

and M_{A^2} has the first homology of a knot complement — torsion-free $H_1 \cong \mathbb{Z}$ — if and only if $m = 1$. All of this depends on A only through $\operatorname{tr} A$.

Proof. Since $\det A = -1$ and $\operatorname{tr} A = m$, the characteristic polynomial is $\chi_A(x) = x^2 - mx - 1$, so Cayley–Hamilton gives $A^2 - mA - I = 0$, that is, $A^2 - I = mA$. Taking determinants, $\det(A^2 - I) = \det(mA) = m^2 \det A = -m^2$.

For the homology, the Wang sequence of the mapping torus of a homeomorphism inducing B on H_1 of the fibre gives $H_1(M_B; \mathbb{Z}) \cong \mathbb{Z} \oplus \operatorname{coker}(B - I)$. Here $B - I = A^2 - I = mA$, and $A \in \mathrm{GL}(2, \mathbb{Z})$ is invertible *over* $\mathbb{Z}$, so $A\mathbb{Z}^2 = \mathbb{Z}^2$ and hence $(mA)\mathbb{Z}^2 = m\mathbb{Z}^2$. Therefore

$$\operatorname{coker}(A^2 - I) \;=\; \mathbb{Z}^2/(mA)\mathbb{Z}^2 \;=\; \mathbb{Z}^2/m\mathbb{Z}^2 \;\cong\; \mathbb{Z}/m \oplus \mathbb{Z}/m,$$

which is trivial precisely when $m = 1$. Every quantity named depends only on $m = \operatorname{tr} A$. □

Remark 4.2 (why this is the right proof)**.** The determinant computation alone gives only $|\operatorname{coker}| = m^2$, which leaves the invariant factors undetermined — on that information the torsion could a priori be cyclic $\mathbb{Z}/m^2$. The identity $A^2 - I = mA$ settles them at once, exhibiting the cokernel as $\mathbb{Z}^2/m\mathbb{Z}^2$ on the nose. Unimodularity of A is what does the work: it is exactly what permits cancelling A from mA without changing the quotient.

Corollary 4.3 (the objection is answered). *The selection of $m = 1$ is constant on* $\mathrm{GL}(2, \mathbb{Z})$*-conjugacy classes of a given trace, and indeed on the entire set of matrices of that trace on the locus. It therefore does not depend on which representative of the class one writes down, and in particular is unaffected by the non-uniqueness of Theorem 3.2 for $m \geq 6$.*

Scope 4.4 (what the identity is, and is not, sensitive to). The identity is specific to the locus: for $\det A = +1$ one has $\chi_A(x) = x^2 - tx + 1$, so $\chi_A(1)\chi_A(-1) = (2-t)(2+t) = 4 - t^2$, and $\det(A^2 - I) = 4 - t^2 \neq -t^2$. A test that passed on both loci would be testing arithmetic rather than the hypothesis; this one separates them. The verification script in Appendix A carries that separation as an explicit control that must fail off the locus.

5 Pricing the period-one restriction

Period one is a restriction. A reader is entitled to ask what it costs. We can answer against the first relaxation, which is period two; we cannot answer in general, and we say so.

Proposition 5.1 (the period-two family). *For $a, b \geq 1$ let*

$$M(a, b) = \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} b & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} ab + 1 & a \\ b & 1 \end{pmatrix}$$

be the matrix of a purely periodic continued fraction of period two. Then:

(i) $\det M(a, b) = +1$ already*: a period-two word is orientation-preserving without squaring;*

(ii) $M(m, m) = \phi_m = X_m^2$*, so the period-one family, squared, is exactly the diagonal of the period-two family — which is therefore strictly larger; and*

(iii) $H_1\big(M_{M(a,b)}; \mathbb{Z}\big) \cong \mathbb{Z} \oplus \mathbb{Z}/\gcd(a, b) \oplus \mathbb{Z}/\operatorname{lcm}(a, b)$*, so the only member of the whole two-parameter family with the homology of a knot complement is $(a, b) = (1, 1)$ — the golden one.*

Proof. (i) and (ii) are immediate from the displayed product. For (iii), $M(a, b) - I = \left(\begin{smallmatrix} ab & a \\ b & 0 \end{smallmatrix}\right)$ has entry gcd $\gcd(a, b)$ and determinant $-ab$, so its invariant factors are $\gcd(a, b)$ and $ab/\gcd(a, b) = \operatorname{lcm}(a, b)$; apply the Wang sequence as above. The torsion vanishes iff $\gcd(a, b) = \operatorname{lcm}(a, b) = 1$, i.e. $a = b = 1$. $\square$

Scope 5.2 (the price, stated exactly). Against the period-two relaxation — the only one we tested — the price of restricting to period one is *zero*: enlarging the family does not produce a competing knot complement. We have not tested period ≥ 3, and we do not assert that the pattern continues. What we assert is that the restriction is not doing hidden work at the first place one would look for it.

6 Summary, and the boundary

The period-one locus is cut out by $\det = -1$ and stratified by trace (Theorem 2.2). The companion representative is canonical exactly up to $m = 5$, with $m = 6$ the first repetition (Theorem 3.2). The homological invariant that picks out $m = 1$ is a function of the trace, hence blind to that ambiguity (Theorem 4.1, Corollary 4.3). Relaxing period one to period two enlarges the family and changes nothing about which member is a knot complement (Proposition 5.1).

The boundary is where §1.3(a) put it. Everything above is a statement about matrices, traces and the homology of mapping tori. The step from a substitution rule to a specific oriented 3-manifold passes through choices — letters to Dehn twists, the puncture, the orientation — that are typed data supplied from outside. A reader who wants the 3-manifold conclusion should regard that step as an assumption, and this paper as settling only what happens once it is granted.

A Verification

Every numeral in this paper is recomputed by `verify/check_locus.py`, which runs 15 checks in four groups and exits non-zero on any failure. Each group carries at least one *control* — a check designed so that a plausible error would make it fail — because a verification suite in which nothing can fail verifies nothing.

group	discharges	content
C1	Thm 2.2	$\lambda_m = m + 1/\lambda_m$ to 50 digits, $m \leq 40$; $(\mathrm{tr}, \det)(X_m) = (m, -1)$; control that the comparison $\det = +1$ family is live
C2	Thm 4.1	$A^2 - I = mA$, $\det(A^2 - I) = -m^2$, and Smith form (m, m) — verified on all 896 integer matrices with $\det = -1$ in a bounded box, traces 1–26, *not* only on X_m; control that the identity *fails* for $\det = +1$; torsion trivial only at $m = 1$
C3	Thm 3.2	primitive class counts at $m^2 + 4$ for $m \leq 11$ by independent reduction and ρ-cycle enumeration; first repetition at $m = 6$; uniqueness for $m \leq 5$
C4	Prop 5.1	$\det M(a, b) = +1$; $M(m, m)$ against the closed form ϕ_m; torsion $(\gcd, \mathrm{lcm})$; only $(1, 1)$ is a knot complement for $a, b \leq 24$

The C2 group is the one that matters. It does not check the identity on the family's chosen representatives; it enumerates every integer matrix of determinant -1 in a box and checks all of them, which is what makes Corollary 4.3 a verified statement rather than a restatement of the algebra.

B Status in the programme register

This paper is the first of four drawn from a larger register of questions, each carrying a status of `PROVED`, `REFUTED`, `CONDITIONAL`, `EXTERNAL_BLOCKER` or `EMPIRICAL`. We record the register entries this paper bears on, including — in fact especially — the ones it does not close.

entry	status	relation to this paper
`OA-C0001`	`REFUTED`	whether minimal description selects a unique seed independently of encoding. Not addressed here; recorded as refuted in the register.
`OA-C0002`	`CONDITIONAL`	whether the declared substitution rules select the Fibonacci rule at minimality. Not addressed here.
`OA-C0003`	`CONDITIONAL`	whether the substitution canonically determines the oriented mapping torus. §1.3(a) is this paper's statement of the same limit.

None of the three is `PROVED`, and this paper does not change that. What it supplies is the part of the chain that *is* unconditional — the characterization of the locus and the trace-only selection along it — while leaving the typed-data step exactly as conditional as the register already records it. We consider stating this more useful than a stronger-sounding framing that the register would not support.

References

[1] O. Perron, *Die Lehre von den Kettenbrüchen*, Teubner, 1913.

[2] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford, 2008, Ch. X.

[3] C. G. Latimer and C. C. MacDuffee, A correspondence between classes of ideals and classes of matrices, *Ann. of Math.* **34** (1933), 313–316.

[4] O. Taussky, On a theorem of Latimer and MacDuffee, *Canad. J. Math.* **1** (1949), 300–302.

[5] D. A. Buell, *Binary Quadratic Forms*, Springer, 1989.

[6] W. P. Thurston, *The Geometry and Topology of Three-Manifolds*, Princeton lecture notes, 1980.